

This is unreleased documentation for Serverpod **Next** version.

For up-to-date documentation, see the [latest version](#) (4.0.0).

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SETUP

 Copy as Markdown

Set up Microsoft sign-in

To set up **Sign in with Microsoft**, you must create an app registration on [Microsoft Entra ID](#) (formerly [Azure AD](#)) and configure your Serverpod application accordingly.

CAUTION

Install the authentication module before you continue. See [Setup](#).

Create your Microsoft Entra ID app

1. Go to [Microsoft Azure Portal](#) and log in with your Microsoft account.
2. Navigate to **Microsoft Entra ID** from the portal menu.



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Manage Microsoft Entra ID
Manage access, set smart policies, and enhance security with Microsoft Entra ID.
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3. Go to **App registrations** and click **New registration**.

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Home

App registrations

[+ New registration](#) [Endpoints](#) [Troubleshoot](#) [Refresh](#) [Download](#) [Preview features](#) [Got feedback?](#)

i Starting June 30th, 2020 we will no longer add any new features to Azure Active Directory Authentication Library (ADAL) and Azure Active Directory Graph. support and security updates but we will no longer provide feature updates. Applications will need to be upgraded to Microsoft Authentication Library (MS

[All applications](#) [Owned applications](#) [Deleted applications](#) [Deactivated applications](#) [Applications from personal account](#)

[Add filters](#)

4. Fill in the required fields:

- **Name:** Enter a name for your application (e.g., "MyApp Authentication").
- **Supported account types:** Choose one of the following:
 - Single tenant only - Default Directory
 - Multiple Entra ID tenants
 - Any Entra ID Tenant + Personal Microsoft accounts
 - Personal accounts only

- **Redirect URI (optional):** Leave this blank for now. We'll add platform-specific URIs later.

The screenshot shows the 'Register an application' page in the Microsoft Azure portal. The header includes the Microsoft Azure logo, a search bar, and a Copilot button. The breadcrumb trail is 'Home > App registrations'. The main heading is 'Register an application'. Below this, there is a section for 'Name' with a red asterisk, a text input field, and a description: 'The user-facing display name for this application (this can be changed later)'. The next section is 'Supported account types', with a description 'Choose the account types that can use this application or access this API', a dropdown menu set to 'Single tenant only - Default Directory', and a 'Help me choose' link. The 'Redirect URI (optional)' section follows, with a description 'We'll return the authentication response to this URI after successfully authenticating the user. Providing this now is optional and it can be changed later, but a value is required for most authentication scenarios.' It contains a 'Select a platform' dropdown and a text input field with the placeholder 'e.g. https://example.com/auth'. At the bottom, there is a note about registering apps and a link to 'Enterprise applications'. A footer line states 'By proceeding, you agree to the Microsoft Platform Policies' with an external link icon. A blue 'Register' button is at the bottom left.

Home > App registrations

Register an application

*** Name**

The user-facing display name for this application (this can be changed later).

Supported account types

Choose the account types that can use this application or access this API

Single tenant only - Default Directory

[Help me choose](#)

Redirect URI (optional)

We'll return the authentication response to this URI after successfully authenticating the user. Providing this now is optional and it can be changed later, but a value is required for most authentication scenarios.

Select a platform

e.g. https://example.com/auth

Register an app you're working on here. Integrate gallery apps and other apps from outside your organization by adding from [Enterprise applications](#).

By proceeding, you agree to the [Microsoft Platform Policies](#)

Register

5. Click **Register** to create your app.

Get app credentials

Get the client ID

After registration, you'll be redirected to the app overview page where you can find your **Application (client) ID**. Copy this value - you'll need it for server configuration.

Microsoft Azure

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Home > App registrations

Test Oauth with Serverpod

Search

Deactivate Delete Endpoints Preview features

Overview

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- Integration assistant
- Diagnose and solve problems
- Manage
 - Branding & properties
 - Authentication (Preview)
 - Certificates & secrets
 - Token configuration
 - API permissions
 - Expose an API
 - App roles
 - Owners
 - Roles and administrators
 - Manifest
- Support + Troubleshooting

Essentials

Display name
[Test Oauth with Serverpod](#)

Application (client) ID
bd059ac1-f9d7-4c78-9885-fabe03496cb5

Object ID
28388a85-47c6-4298-957e-7144c71a1de3

Directory (tenant) ID
14056a5f-84dd-4107-a4f5-e62413dc2cba

Supported account types
[All Microsoft account users](#)

Client credentials
[0 certificate, 1 secret](#)

Redirect URIs
[2 web, 0 spa, 3 public client](#)

Application ID URI
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Managed application in local directory
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State
Activated

Get Started Documentation

Build your application with the Microsoft identity platform

The Microsoft identity platform is an authentication service, open-source libraries, and application management tools. You can create modern, standards-based authentication solutions, access and protect APIs, and add sign-in for your users and customers. [Learn more](#)

Create a client secret

1. In your app's menu, navigate to **Certificates & secrets**.
2. Under **Client secrets**, click **New client secret**.

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Home > App registrations > Test Oauth with Serverpod

Test Oauth with Serverpod | Certificates & secrets

Search

Got feedback?

Overview

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Credentials enable confidential applications to identify themselves to the authentication service when receiving tokens at a web addressable location (using an HTTPS scheme). For a higher level of assurance, we recommend using a certificate (instead of a client secret) as a credential.

Application registration certificates, secrets and federated credentials can be found in the tabs below.

Certificates (0) **Client secrets (1)** Federated credentials (0)

A secret string that the application uses to prove its identity when requesting a token. Also can be referred to as application password.

[+ New client secret](#)

Description	Expires	Value	Secret ID
Client Test Microsoft Oauth in Serv...	11/02/2027	wg1*****	2b61d5da-32ec-4516-934b-6992...

3. Add a description (e.g., "Serverpod Authentication") and choose an expiration period.

4. Click **Add** and immediately copy the **Value** (not the Secret ID). This is your **Client Secret**.

⚠ WARNING

The client secret value is only shown once. Store it securely immediately after creation. Never commit this value to version control.

Get the tenant ID (optional)

If you're restricting authentication to a specific tenant, you'll need your **Directory (tenant) ID**, which is also shown on the app overview page. For most applications, you can use one of these common values:

- **common**: Allows both personal Microsoft accounts and work/school accounts (default).
- **organizations**: Allows only work/school accounts.
- **consumers**: Allows only personal Microsoft accounts.

Configure redirect URIs

You need to configure redirect URIs for each platform you want to support.

1. In your app's menu, navigate to **Authentication**.
2. Click **Add a platform** to configure platform-specific settings.

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information. The left sidebar shows the 'Test Oauth with Serverpod | Authentication (Preview)' page, with the 'Authentication (Preview)' tab selected. The main content area displays the 'Redirect URI configuration' section, which includes a red box around the 'Add Redirect URI' button. A modal dialog titled 'Select a platform to add redirect URI' is open on the right, showing options for 'Web applications' (Web, Single-page application) and 'Mobile and desktop applications' (iOS / macOS, Android, Mobile and desktop applications). The 'Web' option is selected under 'Web applications'.

Web

1. Select **Web** as the platform.
2. Add your redirect URI, typically in the format: `https://yourdomain.com/auth.html`

Add Redirect URI



Web platform

[← All platforms](#) [⚡ Quickstart](#) [📖 Docs](#)

Redirect URI

e.g. `https://example.com/auth`

Web and SPA settings

 Web and SPA settings have been configured. To view or change, please visit the Settings tab.

3. Click **Configure**.

iOS and macOS

1. Select **iOS / macOS** as the platform.
2. For the Bundle ID, enter your app's bundle identifier (e.g., `com.yourcompany.yourapp`).

Add Redirect URI



iOS / macOS platform

[← All platforms](#) [⚡ Quickstart](#) [📖 Docs](#)

Bundle ID

Redirect URI

`com.contoso.myapp`



3. Click **Configure**.

 **NOTE**

For iOS/macOS, Microsoft will automatically generate the redirect URI based on your bundle ID. You can also add custom redirect URIs as needed.

Android

1. Select **Android** as the platform.
2. Enter your Package name (e.g., `com.yourcompany.yourapp`).
3. Enter your Signature hash. You can get this by running:

```
keytool -exportcert -alias SIGNATURE_ALIAS -keystore PATH_TO_KEYSTORE | openssl  
sha1 -binary | openssl base64
```

Add Redirect URI

Android platform

[← All platforms](#) [⚡ Quickstart](#) [📖 Docs](#)

Package name	Signature hash	Redirect URI
com.contoso.myapp	<input "="" type="text" value="2pmj9i4rSx0yEb/viWBYkE/ZQrk="/>	

Generating a development Signature Hash. This will change for each development environment.

On Mac OS,

```
keytool -exportcert -alias androiddebugkey -keystore ~/.android/debug.keystore | openssl sha1 -binary | openssl base64
```

On Windows,

```
keytool -exportcert -alias androiddebugkey -keystore %HOMEPATH%\android\debug.keystore | openssl sha1 -binary | openssl...
```

For more help Signing your apps, check out [Signing your Android app](#)

Generating a production Signature Hash.

4. Click **Configure**.

TIP

For development, use your debug keystore signature hash. For production builds, use your release keystore signature hash. You can add both to Microsoft Entra ID during setup.

Server-side configuration

Store the credentials

Add your Microsoft credentials to the `config/passwords.yaml` file, or set them as environment variables `SERVERPOD_PASSWORD_microsoftClientId`, `SERVERPOD_PASSWORD_microsoftClientSecret`, and `SERVERPOD_PASSWORD_microsoftTenant`.

```
development:
  microsoftClientId: 'YOUR_MICROSOFT_CLIENT_ID'
  microsoftClientSecret: 'YOUR_MICROSOFT_CLIENT_SECRET'
  microsoftTenant: 'common' # or 'organizations', 'consumers', or your specific tenant ID
```

WARNING

Keep your Client Secret confidential. Never commit this value to version control. Store it securely using environment variables or secret management.

Configure the Microsoft identity provider

In your main `server.dart` file, configure the Microsoft identity provider:

```
import 'package:serverpod_auth_idp_server/core.dart';
import 'package:serverpod_auth_idp_server/providers/microsoft.dart';

import 'src/generated/serverpod.dart';

void run(List<String> args) async {
  final pod = Serverpod(args);

  pod.initializeAuthServices(
    tokenManagerBuilders: [
      JwtConfigFromPasswords(),
    ],
    identityProviderBuilders: [
      MicrosoftIdpConfig(
        clientId: pod.getPassword('microsoftClientId')!,
        clientSecret: pod.getPassword('microsoftClientSecret')!,
        tenant: pod.getPassword('microsoftTenant') ?? 'common',
      ),
    ],
  );

  await pod.start();
}
```

TIP

You can use `MicrosoftIdpConfigFromPasswords()` to automatically load credentials from `config/passwords.yaml` or the `SERVERPOD_PASSWORD_microsoftClientId`, `SERVERPOD_PASSWORD_microsoftClientSecret`, and `SERVERPOD_PASSWORD_microsoftTenant` environment variables:

```
identityProviderBuilders: [  
  MicrosoftIdpConfigFromPasswords(),  
],
```

Expose the endpoint

Create an endpoint that extends `MicrosoftIdpBaseEndpoint` to expose the Microsoft authentication API:

```
import 'package:serverpod_auth_idp_server/providers/microsoft.dart';  
  
class MicrosoftIdpEndpoint extends MicrosoftIdpBaseEndpoint {}
```

Generate and migrate

Finally, start the server with `serverpod start` to generate the client code, then create and apply the migration that initializes the database for the provider (in the `serverpod start` terminal, press **M**). More detailed instructions can be found in the general [identity providers setup section](#).

Basic configuration options

- `clientId`: Required. The Application (client) ID of your Microsoft Entra ID app.
- `clientSecret`: Required. The Client Secret generated for your Microsoft Entra ID app.
- `tenant`: Optional. Defaults to `'common'`. Can be `'common'`, `'organizations'`, `'consumers'`, or a specific tenant ID.

For more details on configuration options, see the [customizations page](#).

Client-side configuration

Add the `serverpod_auth_idp_flutter` package to your Flutter app. The Microsoft provider uses `flutter_web_auth_2` to handle the OAuth2 flow, so any documentation there should also apply to this setup.

iOS and macOS

There is no special configuration needed for iOS and macOS for "normal" authentication flows. However, if you are using **Universal Links** on iOS, they require redirect URIs to use **https**. Follow the instructions in the `flutter_web_auth_2` documentation.

Android

In order to capture the callback URL, add the following activity to your `AndroidManifest.xml`. Replace `YOUR_CALLBACK_URL_SCHEME_HERE` and `YOUR_CALLBACK_URL_HOST_HERE` with your actual callback URL scheme and host registered in your Microsoft Entra ID app.

```

<manifest>
  <application>

    <activity
      android:name="com.linusu.flutter_web_auth_2.CallbackActivity"
      android:exported="true"
      android:taskAffinity="">
      <intent-filter android:label="Microsoft Authentication">
        <action android:name="android.intent.action.VIEW" />
        <category android:name="android.intent.category.DEFAULT" />
        <category android:name="android.intent.category.BROWSABLE" />
        <data
          android:scheme="YOUR_CALLBACK_URL_SCHEME_HERE"
          android:host="YOUR_CALLBACK_URL_HOST_HERE" />
      </intent-filter>
    </activity>

  </application>
</manifest>

```

Web

Web sign-in needs the shared callback page that hands the OAuth2 result back to your app. Set it up once as described in [Web callback page \(auth.html\)](#), and point your redirect URI at it, for example <https://yourdomain.com/auth.html>. The same page serves every provider that uses the OAuth2 flow.

Present the authentication UI

Initializing the `MicrosoftSignInService`

Before presenting any sign-in UI, initialize the Microsoft sign-in service. This step is necessary to configure the service with your Microsoft app credentials.

```

await client.auth.initializeMicrosoftSignIn(
  clientId: 'YOUR_MICROSOFT_CLIENT_ID',
  redirectUri: 'YOUR_REGISTERED_REDIRECT_URI',
);

```

INFO

For more information on configuration options and environment variables, see the [customizations page](#).

Using the `MicrosoftSignInWidget`

If you have configured the `SignInWidget` as described in the [setup section](#), the Microsoft identity provider will be automatically detected and displayed in the sign-in widget.

You can also use the `MicrosoftSignInWidget` to include the Microsoft authentication flow in your own custom UI.

```
import 'package:serverpod_auth_idp_flutter/serverpod_auth_idp_flutter.dart';

MicrosoftSignInWidget(
  client: client,
  onAuthenticated: () {
    // Do something when the user is authenticated.
    //
    // NOTE: You should not navigate to the home screen here, otherwise
    // the user will have to sign in again every time they open the app.
  },
  onError: (error) {
    // Handle errors
    ScaffoldMessenger.of(context).showSnackBar(
      SnackBar(content: Text('Error: $error')),
    );
  },
)
```

The widget automatically handles:

- Microsoft sign-in flow for iOS, Android, web, and macOS.
- OAuth2 authentication flow.
- Token management.
- Underlying OAuth2 package error handling.

For details on how to customize the Microsoft sign-in UI in your Flutter app, see the [customizations page](#).